

DSAA3073: Theories in Data Science

Week 5 Validation Sheet: Cross-Validation & Model Selection Criteria

Total Points: 130

Instructions:

- Write your answers clearly and legibly.
- Show all working for full credit.
- State any assumptions you make.
- For proofs, justify each step rigorously.
- This is a practice validation sheet. AI tools and references are allowed.
- Use this to assess your understanding before the actual test.

Part A: Multiple Choice (40 points)

Each question is worth 5 points. Select the single best answer.

Question 1. [5 marks] In the validation set approach, you split data 80/20 for train/validation. How does the validation error relate to the test error of a model trained on all 100% of data?

- A) Validation error equals test error.
- B) Validation error typically underestimates test error.
- C) Validation error typically overestimates test error.
- D) The relationship depends on model complexity.

Question 2. [5 marks] For linear regression with hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, what does $\text{tr}(\mathbf{H})$ equal?

- A) n (number of samples)
- B) d (number of features/parameters)
- C) 1 (always)
- D) The condition number of $\mathbf{X}$

Question 3. [5 marks] Why does 10-fold CV typically have lower variance than LOOCV?

- A) 10-fold CV uses more data for training.
- B) 10-fold CV's training sets overlap less, leading to less correlated validation errors.
- C) 10-fold CV is computationally cheaper.
- D) 10-fold CV has lower bias.

Question 4. [5 marks] What is the key assumption underlying the AIC penalty of $2k$?

- A) The true model is in the candidate set.
- B) The sample size n is small relative to parameters k .
- C) Regular parametric likelihood conditions hold (at least locally).
- D) The data is Gaussian distributed.

Question 5. [5 marks] Which statement about BIC is TRUE?

- A) BIC has a smaller penalty than AIC for all sample sizes.
- B) BIC is consistent: it selects the true model as $n \rightarrow \infty$ when the true model is in the candidate set.
- C) BIC is derived from minimizing KL divergence.
- D) BIC always gives better prediction performance than AIC.

Question 6. [5 marks] The hat matrix for ridge regression is $\mathbf{H}_\lambda = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top$. What happens to $\text{tr}(\mathbf{H}_\lambda)$ as $\lambda \rightarrow \infty$?

- A) $\text{tr}(\mathbf{H}_\lambda) \rightarrow n$
- B) $\text{tr}(\mathbf{H}_\lambda) \rightarrow d$
- C) $\text{tr}(\mathbf{H}_\lambda) \rightarrow 0$
- D) $\text{tr}(\mathbf{H}_\lambda) \rightarrow 1$

Question 7. [5 marks] In nested cross-validation with 5 outer folds and 5 inner folds, how many total model fits are required if you're comparing 10 hyperparameter candidates?

- A) 50
- B) 250
- C) 255
- D) 500

Question 8. [5 marks] Which model selection criterion explicitly penalizes model complexity using $\log n$?

- A) AIC
- B) Training error
- C) LOOCV
- D) BIC

Part B: Mixed Format Questions (90 points)

Question 9. [15 marks] Explain the computational shortcut for LOOCV in linear models.

- (a) [5 marks] Why is naive LOOCV expensive? State the computational complexity for linear regression with n samples and d features.
- (b) [10 marks] Explain the key insight that allows computing LOOCV from a single fit. What role does the hat matrix play? Why doesn't this shortcut work for nonlinear models like neural networks?

Question 10. [13 marks] Compare validation set, 5-fold CV, and LOOCV for hyperparameter selection.

- (a) [6 marks] For each method, describe its bias (as an estimator of full-data test error) and variance. Which has the best bias-variance tradeoff?
- (b) [7 marks] You have $n = 200$ samples for logistic regression with 20 candidate regularization values. For each method, estimate the number of model fits required and recommend which to use. Consider both statistical and computational factors.

Question 11. [15 marks] Derive the AIC formula for Gaussian linear regression.

- (a) [8 marks] Starting from the Gaussian log-likelihood $\log \mathcal{L}(\beta, \sigma^2)$ for $y_i \sim \mathcal{N}(x_i^\top \beta, \sigma^2)$, derive the log-likelihood at the MLEs $\hat{\beta}$ and $\hat{\sigma}^2 = \frac{\text{RSS}}{n}$.
- (b) [7 marks] Apply the AIC formula $\text{AIC} = -2 \log \mathcal{L}(\hat{\theta}) + 2k$ to show that AIC for Gaussian regression simplifies to $\text{AIC} \approx n \log\left(\frac{\text{RSS}}{n}\right) + 2k$ (ignoring constants).

Question 12. [12 marks] Explain selection bias in cross-validation.

- (a) [5 marks] What is selection bias? Why does it occur when using CV for both hyperparameter selection and performance reporting?
- (b) [7 marks] Give a concrete example with ridge regression: suppose you try 10 λ values using 5-fold CV and the CV errors range from 0.20 to 0.25. You select λ^* with CV error 0.20. Explain why reporting “test error = 0.20” is biased. What is the correct procedure?

Question 13. [10 marks] Compare AIC and BIC for a specific scenario.

- (a) [5 marks] For $n = 50$ samples, comparing models with $k = 3$ vs. $k = 6$ parameters, compute the penalty difference for AIC and BIC. Which criterion more strongly favors the simpler model?
- (b) [5 marks] Suppose the true model has 4 parameters, and you're comparing models with 2, 4, 6, and 8 parameters. Which criterion (AIC or BIC) will consistently select the true model as $n \rightarrow \infty$? Explain the asymptotic behavior.

Question 14. [15 marks] Practical model selection scenario.

- (a) [8 marks] You have $n = 150$ samples and want to compare 5 models: linear regression with $d \in \{5, 10, 15, 20, 25\}$ features. Recommend a model selection strategy considering: (i) statistical rigor (avoiding bias), (ii) computational cost, (iii) your goal is prediction. Compare validation set, 10-fold CV, LOOCV, AIC, and BIC.
- (b) [7 marks] After selecting a model using your recommended strategy, how should you report its performance? Should you use the same CV error used for selection? What about confidence intervals?

— End of Validation Sheet —